Grade 7 Geography

Solved Exercises - Balochistan Textbook Board

Balochistan Textbook Board (BTBB)

Unit 1: Plains and Rivers

Exercise (pages 15-16): Choose the correct answers from the options given below:

1. How much of the world's land area is covered by plains?

- ✓ One-third
- Two-third
- One-fifth
- Two-fifth

2. In the Arctic region, the plains of frozen ground are known as:

- Savana
- ✓ Tundra
- Loess
- Alfisol

3. Temperate grasslands of Asia and eastern Europe are called:

- ✓ Steppe
- Savana
- Southern Down
- Prairies

4. How many types of plains are there on the basis of formation?

- ✓ 2
- 3
- 4

5

5. Nile River is situated in:

✓ Egypt

Brazil

Pakistan

France

Note on MCQ 1: the unit says plains cover more than one-third of the world's land area (page 6) and about one-third of the global land area (page 9); One-third is the intended option. Note on MCQ 4: on the basis of formation the unit groups plains broadly into two types, erosional plains and depositional plains.

Exercise: Briefly answer the following questions:

1. Differentiate among plain, plateau and mountain.

A plain is a flat area of land that is mostly level, with little to no hills or mountains and low relief. A mountain has a high relief with steep slopes. A plateau has a relatively flat or undulating surface raised above the surrounding land. A valley is a low area of land between hills or mountains.

2. What are erosional plains?

Erosional plains are plains formed by the wearing down of land by natural agents such as water, wind or glaciers. Over time these agents wear down the land, smoothing out irregularities and creating a relatively flat surface.

3. What is meant by flood plain?

A flood plain, also called a fluvial plain, is a flat area of land near a river, formed by the erosion and deposition of sediments when the river spreads out and drops sand, mud and other materials. It is very fertile and can flood easily when the river overflows. Examples are the Indus River floodplain in Pakistan, the Ganges-Brahmaputra floodplain in Bangladesh and India, and the Nile River floodplain in Egypt.

4. Enlist the main components of the river system.

The main components of a river system are: source, tributaries, main channel, drainage basin (watershed), river valley, floodplain, and river delta. A tributary is a fresh water channel that feeds the main river.

5. Enlist the landforms made by river in its middle course.

The landforms made by a river in its middle course are meanders (curved bends), ox-bow lakes (cut-off meanders), flood plains (flat, fertile areas), natural levees (raised banks), and braided channels.

Exercise: Give detailed answers of the following questions:

1. Describe the common types of plains.

The common types of plains are: (1) Fluvial plains, also called flood plains, formed by erosion and deposition of sediments during periodic flooding, for example the Indus, Ganges-Brahmaputra and Nile floodplains. (2) Aeolian plains, formed by erosion and deposition by wind over time, consisting of sand and silt, for example the Cholistan and Thar deserts, the Sahara and the Gobi. (3) Glacial plains, formed when melting glaciers release and deposit

the materials they carried, for example the Great Plain of North America and the Central Lowlands of Europe. (4) Coastal plains, located along coastlines and formed by erosion and deposition by waves or tides, for example the Indus River Delta and the Atlantic Coastal Plain. (5) Lacustrine plains, formed by the deposition of sediments in lakes, for example the Kashmir Valley. (6) River plains, flat fertile areas near a river formed when the river spreads out and drops sand, mud and other materials.

2. Explain the three courses of a river.

From source to mouth a river is divided into three courses. Upper course: the starting section, characterised by steep gradients, fast-flowing water and narrow channels; erosion is dominant and creates V-shaped valleys, waterfalls, rapids and gorges. Middle course: also called the mature or valley stage, the transitional phase with a less steep gradient and slower-flowing water; it combines water supply, transportation, energy generation, ecology and recreational value. Lower course: also called the delta or old age stage, the final stage before the river reaches the ocean, a lake or another body of water; here the river mainly deposits sediment rather than eroding, and distributes into channels forming deltas.

3. Why deltas are formed in the lower course of river?

A river delta is a triangle shaped landform formed at the mouth of a river where it enters a body of water such as an ocean or a lake. As a river reaches its lower course, the gradient or slope of the land decreases, so the velocity of the river flow falls. The lower velocity allows the river to deposit the sediments it has been carrying from

the upper and middle courses, and this deposition eventually forms a delta. For example, the Indus River falls in the Arabian Sea and forms a delta.

4. What is the role of river in the economy of a region?

Rivers are crucial for the economy of a region in several ways: they are a crucial source of fresh water for human populations, agriculture and industries; fertile floodplains and deltas formed by rivers provide rich soil for agriculture; river transport is cost-effective and environment friendly, contributing to regional and international trade, economic growth and riverfront industries; rivers with significant water flow and elevation support hydropower generation, a renewable energy source; and rivers and their landscapes attract tourists and offer recreation such as boating, fishing, rafting and wildlife watching, which generate revenue and support local businesses.

5. Enlist the landforms made by the river in its courses.

Upper course: V-shaped valley, waterfalls, rapids and gorges. Middle course: braided channels, meanders, flood plains, ox-bow lakes and levees. Lower course: deltas, distributaries and estuaries.

Guidance (Activity, page 14): assign students to get information about the Indus Plain of Pakistan, one of the most fertile plains in the world, prepare a chart and present it in the class. Guidance (Project, page 14): in small groups, create a model of a river system in a large tray using sand or soil, pebbles and small stones for the banks, plastic cups for mountains or hills, blue-coloured water for

the river, small plants for the riparian zone and toy animals or pictures for river animals, then discuss the animals' dependence on the river for food, water and shelter. These are practical tasks; the book fixes no single written answer for them.

Unit 2: Water Resources and Management

Exercise (pages 32-33): Choose the correct answers from the options given below:

1. In how many categories, the major sources of water on the Earth can be classified?

- two
- ✓ three
- four
- five

2. Amazon River is situated in

- ✓ South America
- North America
- Australia
- Pakistan

3. The largest freshwater lake of the world is

- Lake Mancher
- ✓ Lake Superior
- Lake Eri
- Lake Michigan

4. The most common type of precipitation is

- hail
- sleet

✓ rain

snow

5. Which of the following types of rain is also known as relief rain?

✓ orographic

coastal

convectional

fontal

6. The process by which water vapours are released into the atmosphere by plants is called

condensation

✓ transpiration

precipitation

Sublimation

7. Greener Fuel is another name for

coal

mineral oil

natural gas

✓ biogas

Exercise: Fill in the blanks.

1. ____ occurs when ice or snow turns directly into water vapour without melting into a liquid state.

Sublimation

2. ____ forms when temperature in the atmosphere is below

freezing point

Snow

3. ____ is a method that involves adding chlorine to water to kill bacteria and viruses.

Chlorination

4. The main component of biogas is ____ gas.

methane

5. The water cycle is also known as the ____.

hydrological cycle

Exercise: Briefly answer the following questions.

1. Enlist major sources of water on the Earth.

The major sources of water on the Earth can be classified into three categories: surface water, ground water and atmospheric water.

2. Distinguish between ground and surface water sources.

Surface water is water found on the surface of the Earth in the form of lakes, rivers, streams and reservoirs, including both freshwater and saline water bodies. Ground water is water stored under the Earth's surface; it gets refilled when rainwater and surface water soak into the ground, and it is accessed through wells, springs and aquifers.

3. Define precipitation.

Precipitation refers to any form of water that falls from the

atmosphere to the surface of the Earth. It occurs when water vapours turn into droplets during condensation; these droplets join each other to become heavy and fall downward under gravity.

4. Why is precipitation important in the water cycle?

Precipitation is important in the water cycle because it is a primary source of replenishing both surface water and groundwater. It helps sustain ecosystems, replenish water supplies and regulate the global climate, and it is a vital part of distributing water across the planet.

5. Where does the wastewater go? Write two different ways of using sewage waste.

Wastewater goes through the sewage treatment process: screening, primary treatment, secondary treatment and final treatment, after which the treated water is returned to rivers and the solids to land, and power is generated. Two different ways of using sewage waste are: (1) to irrigate agricultural fields, landscaping and urban green spaces, which conserves freshwater; (2) for industrial purposes such as power plants, manufacturing facilities and cooling systems. Other uses named by the unit are recharging groundwater, non-potable uses in buildings, and recreation in designated areas.

Exercise: Give detailed answers of the following questions.

1. Write about different types of precipitation.

The different types of precipitation described in the unit are: Rain, the most common type, formed when water droplets in the

atmosphere become large enough to fall to the ground; raindrops typically have a diameter between 0.5 and 5 millimetres, and rain includes orographic (relief) rain, convectional rain and frontal rain. Snow, which forms when the temperature in the atmosphere is below freezing point and water vapour turns directly into ice crystals that join to form snowflakes. Sleet, which occurs when raindrops fall through a layer of freezing air near the ground and form a hard outer layer containing cold water inside. Dew, the moisture that forms on surfaces such as grass, leaves and metal objects during the cool hours of the night or early morning when surface temperatures drop below the dew point. Hail, solid ice pellets or balls formed when strong updrafts in thunderstorm clouds carry raindrops upward into extremely cold areas where they freeze and gain layers of ice until they become too heavy and fall.

2. Write the processes of water cycle.

The water cycle, also known as the hydrological cycle, is the continuous process by which water circulates between the surface of the Earth, the atmosphere and back again. Its key processes are: Evaporation, the conversion of liquid water into water vapour which rises into the atmosphere. Transpiration, the process by which plants absorb water through their roots and release water vapour from their leaves into the atmosphere. Condensation, when water vapour in the atmosphere cools and transforms back into liquid water, forming clouds. Sublimation, when ice or snow turns directly into water vapour without melting into a liquid state. Precipitation, when water falls back to the Earth's surface as rain, snow, sleet or hail, replenishing rivers, lakes and groundwater. Runoff, when

precipitation flows over the land into rivers, lakes and oceans. Infiltration, when precipitation seeps into the ground and becomes groundwater in aquifers.

Guidance (Investigation, detailed question 3): carry on an investigation about scarcity of clean drinking water in Pakistan. The unit names the factors behind the problem: population growth, climate change and inefficient water usage practices, and notes that Pakistan faces challenges related to water scarcity and management. This is an open investigation task; the book fixes no single written answer for it, so findings should be recorded from the investigation itself.

4. Justify how biogas is considered as “Greener Fuel”.

Biogas is considered a Greener Fuel because it is a type of green energy produced from organic waste materials such as food scraps, animal manure and plant residues. Its main component is methane gas, which is a clean-burning fuel. Using biogas is more environment friendly because it produces fewer greenhouse gases that contribute to climate change; it reduces dependence on fossil fuels, which are limited and can harm the environment when burned; and it provides a way to manage organic waste effectively and turn it into something useful. It is renewable because the materials used to produce it can be replenished naturally, and it can be used for cooking, heating and even generating electricity.

Unit 3: Settlement and Land Use

Exercise (pages 46-47): Choose the correct answers from the options given below:

1. In the beginning, the man lived in

- ✓ isolution
- settlements
- groups
- communities

2. How many types of settlements are there?

- ✓ two
- three
- four
- five

3. A small and compact settlement is called

- city
- ✓ halmet
- town
- megapolis

4. The city of Karachi is a

- city
- megapolis
- town

✓ metropolitan

5. It refers to an extended urban area formed by merging of adjacent cities or town

primate

megapolis

halmet

✓ conurbation

Note on MCQ 4: Karachi appears in the unit both as a major city (commercial hub) and as a metropolitan-area example (economic powerhouse); metropolitan is ticked as the distinguishing classification, and the dual listing is stated here. Note on MCQ 3 and 5: the options are printed as halmet and primate; they are kept as printed.

Exercise: Fill in the blanks.

1. ____ are small, compact settlements with a limited number of houses.

Hamlets

2. The ____ of a location plays a vital role in the settlement suitability.

topography

3. ____ settlement often functions as centers of commerce and trade.

Urban

4. Settlement primarily serves as places of ____ for people.

residence

5. An extended urban area formed by the merging of adjacent cities or towns is called ____.

conurbation

Exercise: Briefly answer the following questions.

1. Define settlement.

A settlement refers to a community or a place where people reside and engage in various activities, such as living, working and interacting with one another.

2. Write two suitable conditions for the dwelling of settlement.

Two suitable conditions for the dwelling of a settlement are: (1) Availability of water: access to a reliable water source is important, such as a river, lake or underground water source, to meet the needs of the population for drinking, cooking and agricultural purposes. (2) Topography and land: flat or gently sloping land is generally more suitable for construction, agriculture and infrastructure development, while steep slopes, flood-prone areas or areas prone to landslides may pose challenges.

3. Write at least two differences between conurbations and megapolis.

Two differences are: (1) A conurbation is an extended urban area formed by the merging of adjacent cities or towns, occurring when separate urban areas grow and eventually blend together, creating

a continuous built-up area. A megapolis is an even larger and more extensive urban region. (2) A conurbation is characterised by a high population density and a significant level of interaction and interdependence between the constituent urban areas, while a megapolis refers to a vast metropolitan area that encompasses multiple cities, towns and suburbs, often spanning across different administrative boundaries or even multiple states or provinces, with a massive population, extensive infrastructure networks and diverse economic activities.

4. What is a hamlet?

Hamlets are small, compact settlements with a limited number of houses. They are often found in remote or isolated areas, where the population is small and primarily engaged in agriculture or other traditional livelihoods.

5. Differentiate between towns and cities.

Differences between towns and cities given in the unit: Size: towns are relatively smaller in size, cities are larger in size. Population: towns generally have a lower population, cities have a higher population. Infrastructure: towns have basic infrastructure, cities have advanced infrastructure. Economic activities: towns have a limited range of economic activities, cities have a diverse range. Services: towns have limited-service availability, cities have a wide range of services including healthcare, education and entertainment. Transportation: towns have relatively simpler transportation systems, cities have more complex transportation networks including public transportation. Cultural and social

activities: towns have limited cultural and social activities, cities have rich cultural and social activities including museums, theatres and festivals. Opportunities: towns have limited employment and business opportunities, cities have more employment and business opportunities. Housing: towns have a smaller variety of housing options, cities have a wider range. Land use: towns are less urbanized, with more open spaces; cities are more urbanized, with limited open spaces.

Exercise: Give detailed answers of the following questions.

1. Compare and contrast urban and rural settlements.

Rural settlements are characterised by a low population density and are typically located in rural or agricultural areas; they are often associated with farming, agriculture and primary industries, and can take the form of dispersed settlements (houses spread out over a larger area), clustered settlements (houses located close together, often where water or fertile land are concentrated), linear settlements (elongated, following a road, river or coastline) or nucleated settlements (a central core or nucleus, such as public buildings or markets, surrounded by scattered houses or agricultural land). Urban settlements are large, densely populated areas characterised by a high concentration of people, buildings and infrastructure, usually associated with industrial, commercial and service-oriented activities, ranging from small towns to major cities. In short, rural settlements are small, low-density and agriculture-based, while urban settlements are large, high-density and industry, commerce and service-based.

2. The dwelling of settlement needs certain suitable conditions. Identify those conditions.

The suitable conditions for the dwelling of a settlement identified in the unit are: (1) Availability of water: a nearby reliable source such as a river, lake or underground water for drinking, cooking and agriculture. (2) Topography and land: flat or gently sloping land suits construction, agriculture and infrastructure. (3) Suitable climate: moderate and favorable climates with moderate temperatures, sufficient rainfall and suitable growing seasons are preferred; extreme heat, cold or excessive rainfall pose challenges. (4) Natural resources: fertile soil for agriculture, forests for timber and fuel, minerals for economic activities and renewable energy sources. (5) Transportation and accessibility: nearness to roads, railways or waterways and to nearby towns, cities and markets. (6) Security and safety: a reasonable level of security, low crime rates, nearness to emergency services and adequate measures against natural disasters. (7) Social and community factors: a sense of community, access to healthcare facilities, educational institutions, recreational spaces and cultural amenities. (8) Economic opportunities: job opportunities, markets, trade routes and a favourable business environment. Historical and cultural factors may also influence settlement patterns.

3. Write advantages and disadvantages of rural settlement.

Advantages of rural settlements: natural environment, community and social bonds, lower cost of living, reduced noise and pollution, and connection with nature. Disadvantages of rural settlements: limited employment opportunities, lack of infrastructure and

services, limited access to resources, social isolation, and distance from urban centres.

4. Write the functions of settlement.

The functions of a settlement are: (1) Residential function: settlements primarily serve as places of residence, providing housing and infrastructure for individuals, families and communities. (2) Commercial function: settlements often function as centers of commerce and trade, with markets, shops, malls and other commercial establishments where goods and services are bought and sold. (3) Industrial function: some settlements are designated for industrial activities, with factories, manufacturing plants and industrial zones where goods are produced or processed. (4) Administrative function: settlements can serve as administrative centers, hosting government offices, administrative buildings and public institutions. (5) Educational function: settlements have educational institutions such as schools, colleges and universities. (6) Recreational and cultural function: settlements offer spaces for recreation, entertainment and cultural activities, with parks, gardens, theatres, museums and sports facilities. (7) Transport and communication function: settlements serve as transportation and communication centers, with networks such as roads, railways, airports and ports. (8) Agricultural function: in rural settlements, agricultural activities play a significant role, with farmland and agricultural infrastructure where crops are grown, livestock is raised and agricultural products are processed.

Guidance (You Can Do, page 47): find information from various

sources about how much Pakistan is successful in achieving Sustainable Development Goal 11, Sustainable Cities and Communities, and suggest ways to ensure sustainable urban development; and visit a small settlement near your hometown (with teachers or parents), interview people living there about their lifestyle and economic conditions, and suggest ways of improving their living conditions (sanitary conditions, education and economic wellbeing). Guidance (Project, pages 44-45): design a Green and Clean City poster, using drawing paper or poster board, colour pencils, markers or crayons, glue, tape and scissors, showing clean streets, green spaces, recycling bins, renewable energy sources and people engaged in environment friendly activities, then share and display the posters. These are practical and investigation tasks; the book fixes no single written answer for them.

Unit 4: Agriculture

Exercise (pages 66-67): Choose the correct answers from the options given below:

1. Human life initially relied for food on gathering and

- agriculture
- ✓ hunting
- industries
- lumbering

2. Which theory suggests that agriculture arose in response to climate changes?

- river valley theory
- maiden heap theory
- ✓ oasis theory
- lumbering

3. What percentage of global workforce is engaged in agriculture sector

- 35.5
- 36.5
- 37.5
- ✓ 38.5

4. In rural areas of Pakistan, particularly the northern regions, which type of agriculture is commonly practised?

✓ subsistence

intensive

extensive

commercial

5. In this method, multiple crops are grown together in the same field

Monoculture

✓ polyculture

Crop rotation

mixed crop

6. In 2020, the total arable land in Pakistan was

35.12%

✓ 40.12%

45.12%

50.12%

7. How much percent of livestock sector contribute?

✓ 11%

12%

13%

14%

8. Which is the largest crop in term of production in Pakistan?

rice

✓ wheat

cotton

sugarcane

Note on MCQ 3: the unit text (page 50) says almost 38.5 percent of the global workforce is engaged in the agriculture sector. Note on MCQ 5: polyculture is ticked because the unit defines it as growing multiple crops in the same field; mixed cropping (two or more different crops simultaneously on the same piece of land) is closely related and is noted here. Note on MCQ 6: the unit text (page 55) reports total arable land in Pakistan at 40.12 percent in 2020 (World Bank); the Key Points box (page 65) prints almost 42.12 percent instead, which does not match the MCQ options. Note on MCQ 8: the unit states wheat is the largest crop in terms of production in Pakistan (page 58), even though sugarcane tonnage figures in the unit's own table are higher; the book's statement is followed.

Exercise: Fill in the blanks.

1. ____ Types of raw material are obtained from farming.

Two

2. ____ cultivation is an important part of Pakistan's agricultural sector.

Crop

3. The theory that suggests that agriculture arose in response to climate change is called ____.

Oasis Theory

Note on fill 1: the unit divides farming raw materials into two types, crop-based raw materials and livestock-based raw materials. Note

on fill 2: the unit names crop cultivation as a main branch of agriculture; the fill sentence names no single crop, so Crop is the best reading, flagged here rather than presented as the book's own named crop.

Exercise: Briefly answer the following questions.

1. What is Oasis Theory?

The Oasis Theory suggests that agriculture arose in response to climate changes, specifically during the transition from the Ice Age to the present epoch. Global warming led to glacier retreat and the formation of oases in arid regions. These oases attracted plants and animals, providing a reliable food source for humans. Humans settled near these oases, leading to the domestication of plants and the emergence of agriculture.

2. Enlist the types of agriculture.

Based on different factors, agriculture can be broadly categorized into: subsistence agriculture, intensive agriculture, extensive agriculture, commercial agriculture, mixed farming, truck farming, and livestock farming. Its two main branches are crop cultivation and livestock farming.

3. Differentiate between intensive and extensive agriculture.

Intensive agriculture maximizes land productivity through advanced techniques and high inputs; it is practised in densely populated areas with limited arable land, for example high-yielding rice varieties cultivated intensively in the Punjab region of Pakistan. Extensive agriculture is practised in areas with abundant land and limited labor, and includes practices like livestock grazing and

large-scale crop cultivation with low inputs; it is practised in the USA, Canada, Russia and Brazil.

4. Differentiate between polyculture and monoculture.

Polyculture is the practice of growing multiple crops in the same field, usually without a specific spatial arrangement or pattern, involving a diverse range of crops such as grains, vegetables, fruits and herbs without a strict hierarchy or dominant crop. Monoculture is the practice of cultivating a single crop species on a given piece of land over a significant area, often with specialized machinery, inputs and management practices; it can facilitate efficient management and harvesting but can also lead to increased vulnerability to pests, diseases, soil degradation and reduced biodiversity.

5. How Rabi crops are different from Kharif crops?

Rabi crops are sown and harvested during the winter season, which typically starts in October/November and lasts until March/April; they include wheat, barley, grams, mustard and potatoes. Kharif crops are sown during the monsoon summer season, from around June/July until September/October; they are rain-fed and rely on abundant rainfall and warmer temperatures, and include rice, cotton, maize, sugarcane and millet.

Exercise: Give detailed answers of the following questions.

1. Analyze and compare different methods of farming practices in the world.

Traditional and conventional methods include agroforestry

(intentional planting of trees on farms to provide shade, microclimate, windbreaks, erosion control and additional income), crop rotation (growing different crops in a specific sequence on the same piece of land to improve soil health and manage pests), mixed cropping or intercropping (two or more different crops simultaneously on the same piece of land, giving efficient use of resources, better pest and disease management, improved soil health and increased biodiversity), polyculture (multiple crops in the same field without a strict pattern) and monoculture (a single crop species over a significant area). Modern methods include organic agriculture (avoiding synthetic chemicals like pesticides and fertilizers, relying on compost, crop rotation and natural pest control) and sustainable agriculture (farming that meets current needs without harming the environment or depleting resources for future generations, balancing food production with care for soil, water, biodiversity and farming communities).

2. Describe Pakistan as an agricultural country.

Pakistan is an agricultural country with a varied climate, primarily arid, with hot summers and cool to cold winters; irrigation is crucial, with an extensive infrastructure of dams, canals and tube wells, canal irrigation from the Indus River system being the primary water source. Agriculture is considered the backbone of the economy: it contributes about 18.9 percent of Gross Domestic Product, accounts for 42.3 percent of the employed labour force, and is the largest source of foreign exchange earnings. Total arable land was reported at 40.12 percent in 2020, declining due to rapid urbanization. Major crops (wheat, rice, cotton and sugarcane)

contribute around 4.18 percent while minor crops contribute 2.1 percent to GDP. The livestock sector contributes 11 percent to GDP and employs approximately 35 million people; fisheries and forestry each contribute an estimated 0.4 percent. Pakistan is among the world's top ten producers of wheat, cotton, sugarcane, mango, dates and kinnow oranges, and is ranked 10th in rice production. The major agricultural province is Punjab, where wheat and cotton are the most grown, and 60 percent of the agricultural share of exports comes from Punjab.

3. Write in detail the crops grown in Pakistan.

On the basis of seasons, crops are classified into Rabi crops (winter: wheat, barley, grams, mustard, potatoes) and Kharif crops (monsoon summer: rice, cotton, maize, sugarcane, millet). On the basis of purpose, crops are cash crops (cotton, tobacco, sugarcane, tea, oilseeds, grown for commercial value) and fodder crops (alfalfa, maize fodder, berseem, lucerne, grown for feeding livestock). Major crops: Wheat, the largest crop in terms of production and the most important staple, grown in Punjab, Sindh, Balochistan and Khyber Pakhtunkhwa, with average annual production around 25-27 million metric tons. Rice, especially in Punjab and Sindh, including Basmati, around 7-8 million metric tons annually. Cotton, an important cash crop known as Silver Fiber, grown in Punjab and Sindh, around 10-12 million bales of 170 kg annually. Sugarcane, mainly in Punjab and Sindh, grown for sugar, molasses and ethanol, around 60-70 million metric tons annually. Maize, also known as corn, concentrated in Punjab, Khyber Pakhtunkhwa and Sindh, around 5-6 million metric tons annually.

Fruits include mangoes (king of fruits, almost 2 million metric tons in 2022), citrus fruits, apples, bananas, guavas, dates and grapes. Vegetables (potatoes, tomatoes, onions, garlic, cucumbers, okra, carrots, peppers), pulses (lentils, chickpeas, beans) and oilseeds (sunflower, canola, sesame) are also cultivated.

Guidance (Project, pages 64-65): design a sustainable garden on a small plot, considering types of plants, companion planting, efficient water usage (drip irrigation, rainwater harvesting), natural pest control, composting and attracting pollinators, then draw, label and present the design. Guidance (Activities): draw a bar graph of agricultural growth in Pakistan by year from the unit's table and analyse the changes; use the Pakistan Bureau of Statistics, FAO Pakistan and Pakistan Agricultural Research Council websites to find crop production data and draw and analyse a bar graph. These are practical tasks; the book fixes no single written answer for them.

Unit 5: Climate Change

Exercise (pages 80-81): Choose the correct answers from the options given below:

1. Short term atmospheric conditions are called

- ✓ weather
- climate
- biodiversity
- precipitation

2. IPCC stands for

- intergovernmental panel on climate concerns
- international panel on climate change
- international panel on climate concerns
- ✓ intergovernmental panel on climate change

3. The Earth receives heat from

- the mars
- the jupitar
- the moon
- ✓ the sun

4. Ozone layer is located in this layer of atmosphere

- troposphere
- ✓ stratosphere
- mesosphere

thermosphere

5. Ozone depletion is mainly caused by

sunlight

✓ cfcs

uv rays

atomic energy

Note: the printed exercise numbers the ozone-depletion item as 3 a second time; it is given here as item 5 in corrected sequential order. Option spellings jupitar and cfcs are kept as printed.

Exercise: Fill in the blanks.

1. Planting trees and preserving green spaces have numerous ____ benefits.

environmental

2. Conserving energy reduces our ____ and lowers greenhouse gas emissions.

carbon footprint

3. ____ effect is important for life on the Earth because it keeps our planet warm enough for us to survive.

Greenhouse

4. The ____ is a special part of the atmosphere of the Earth that acts like a shield.

ozone layer

Note on fill 4: the printed fill says acts like a sluiced; the unit's own definition (page 74 and Key Points) says the ozone layer acts like a shield, and shield is used here.

Exercise: Briefly answer the following questions.

1. Differentiate between weather and climate.

Weather refers to the short-term atmospheric conditions in a specific region at a given time, including day-to-day variations in temperature, humidity, precipitation, wind speed, cloud cover and atmospheric pressure. Climate refers to the long-term patterns and averages of weather conditions observed in a particular region over an extended period of 30-40 years, including average temperature, precipitation, wind patterns and other atmospheric variables.

2. Define climate change.

Climate change refers to the long-term changes in the Earth's weather patterns and temperatures. It is caused by natural processes and human activities, such as burning fossil fuels (like coal, oil and gas) and cutting down forests, which release gases that trap heat in the atmosphere.

3. What is greenhouse effect?

The greenhouse effect is a natural process in the atmosphere of the Earth. The Earth receives heat from the Sun in the form of sunlight; some of this heat is absorbed by the surface of the Earth such as land, water and trees. When the surface gets heated, it releases some heat radiation. Certain gases in the atmosphere, like carbon dioxide and methane, act like the glass walls of a greenhouse: they trap some of the heat radiated from the surface, warming the

atmosphere and making it a comfortable place for us to live. Human activities such as burning fossil fuels release extra amounts of these gases, causing an imbalance that increases the temperature of the Earth, which is called global warming.

4. What do you know about 3Rs strategy?

The three Rs of sustainability are a fundamental approach to minimize waste and conserve resources: Reduce, Reuse, Recycle. We can reduce waste by opting for reusable items like water bottles, shopping bags and coffee cups; reuse items, such as donating or repurposing clothes and furniture; and recycle materials like paper, plastic and glass to reduce the need for new resources.

5. Enlist the practices of water conservation.

Practices of water conservation given in the unit are: taking shorter showers and turning off the tap while brushing teeth; fixing leaks in faucets or pipes promptly; collecting rainwater for gardening and outdoor use; and using water-efficient appliances and fixtures.

Exercise: Give detailed answers of the following questions.

1. Analyze the changes caused by greenhouse effect.

Changes caused by the greenhouse effect, when disrupted by human activities, are: (1) Global warming: increased greenhouse gases like carbon dioxide raise the temperature of the Earth, causing weather changes, ice melting and ecosystem disruptions; the Earth's average surface temperature has risen by about 1.1 degrees Celsius since the late 19th century, and about 750 billion

tons of ice is melting every year. (2) Climate change: rising global temperatures cause extreme weather like heatwaves, droughts, hurricanes and heavy rainfall, with significant impacts on agriculture, water resources and human settlements. (3) Rising sea levels: melting ice sheets and glaciers add water to oceans, threatening coastal areas with erosion, flooding and displacing coastal communities. (4) Extinction of species: loss of biodiversity affects ecosystem health and functioning. (5) Impact on agriculture: changes in temperature, rainfall patterns and extreme weather affect agriculture; shifts in seasons, water availability and pests can reduce crop yields, disrupt food production and impact food security and the economy. (6) Health effects: heatwaves and high temperatures cause heat-related illnesses and stress, and climate change favours disease-carrying organisms like mosquitoes, increasing the risk of vector-borne diseases such as dengue fever.

2. What is Ozone layer? Briefly describe the causes of ozone depletion.

The ozone layer is a special part of the atmosphere of the Earth that acts like a shield. It is located in the stratosphere at about 10 to 50 kilometers above us, and is made up of ozone gas, three oxygen atoms (O3) bonded together. It protects us from the harmful ultraviolet (UV) rays of the sun, which can cause sunburn, skin cancer, eye diseases and other problems. Causes of ozone depletion: it is mainly caused by human-made chemicals called chlorofluorocarbons (CFCs) and other ozone-depleting substances, commonly used in aerosol sprays, air conditioners, refrigerators and foam insulation. When released, they rise to the ozone layer,

are broken down by sunlight, and release chlorine and bromine atoms that destroy ozone molecules; one chlorine or bromine atom can destroy thousands of ozone molecules. The Montreal Protocol, an international agreement signed by many countries, was established to limit the production and use of ozone-depleting substances, and the ozone layer is slowly recovering.

3. Identify and explain the main causes of climate change.

The main causes of climate change are the greenhouse effect and the resultant global warming. Greenhouse gases such as carbon dioxide, methane and nitrous oxide act like a blanket or shield, trapping some heat instead of letting it escape into space and making the Earth a habitable planet. Human activities such as burning fossil fuels (coal, oil and gas) for energy, deforestation and certain industrial processes have increased the amount of greenhouse gases released into the atmosphere, thickening the blanket and trapping more heat. Global warming is the main driver of climate change, bringing rising temperatures, melting ice caps and glaciers, rising sea levels, changes in precipitation patterns, and more frequent and severe weather events like storms and heatwaves.

4. Write in detail some of the practices that we can adopt for our green future.

Practices we can adopt for a green future, from the unit's Go Green section: (1) Reduce, Reuse, Recycle: use reusable items, donate or repurpose clothes and furniture, and recycle paper, plastic and glass. (2) Conserve energy: turn off lights and appliances when not

in use, use energy-efficient bulbs and appliances, adjust thermostats, unplug chargers and electronics, and use natural light. (3) Use renewable energy sources: install solar panels, use wind turbines, and support policies that promote renewable energy. (4) Practise water conservation: shorter showers, turn off the tap while brushing teeth, fix leaks promptly, collect rainwater, and use water-efficient appliances. (5) Plant trees and support green spaces: join tree planting initiatives, create gardens at home or school, and support reforestation organisations. (6) Choose sustainable transportation: walk or cycle for short distances, use public transportation, carpool, and choose electric or hybrid vehicles when possible. (7) Support local and sustainable products, such as organic food or environment friendly household items.

Guidance (Activities, page 79): Pakistan is the 6th most vulnerable country to climate change — find reasons and their possible impacts and suggest ways to overcome them, using the Climate Change Division, Global Change Impact Studies Centre, World Bank and UNDP sources named in the book; and solve the word puzzle printed in the book. These are research tasks; the book fixes no single written answer for them.

Unit 6: Minerals and Power Resources

Exercise (pages 94-95): Choose the correct answers from the options given below:

1. Iron ore, copper, gold, silver, lead and zinc are minerals

- ✓ metallic
- non-metallic
- non-ferrous
- energy minerals

2. A traditional mining method of manually extracting gold is

- open-cast
- adit
- shaft
- ✓ hand paining

3. In Pakistan, one of the largest untapped coal deposits in the world found is

- Dandot
- Paddh
- ✓ Thar
- Makarwal

4. The highest carbon contents are found in

- peat
- lignite

✓ bituminous

chilghazi

5. One of the largest iron deposits in Pakistan is

Langrial

✓ Kalabagh

Qrakaram

Chilghazi

6. Bhikki Thermal Power Plant is situated in

Guddu

Sahiwal

Gujranwala

✓ Shekhupura

Note on MCQ 2: the printed option is hand paining; the unit's own method name is Hand-Panning Mining, a traditional method of manually extracting gold from rivers and streams using a shallow pan. Note on MCQ 4: the unit states anthracite has the highest carbon content (86 percent to 97 percent), but anthracite is not among the printed options; bituminous, at 70 percent to 86 percent carbon, is the highest-carbon option listed and is ticked on that basis. Note on MCQ 5: the unit says the Kalabagh iron deposit is one of the largest in the country, with measured reserves of 350 million tonnes.

Exercise: Fill in the blanks.

1. Based on the carbon content, there are ____ types of coal.

four

2. ____ is used in construction, electrical wiring, coins and electronics.

Copper

3. Fossils fuels including ____, ____ and ____ are commonly used to generate power in various industries and for electricity Production.

coal, oil and natural gas

4. Pakistan has extensive salt reserves, particularly in ____, one of largest and oldest mines in the world.

Khewra

Note on fill 3: the three blanks are filled together (coal, oil and natural gas) because the builder renders one filling word per item; the unit's fossil-fuel list is coal, oil and gas.

Exercise: Briefly answer the following questions.

1. Enlist different types of minerals.

Different types of minerals are: (1) Metallic minerals, which contain metallic elements, have a lustrous appearance and are good conductors of heat and electricity, for example iron ore, copper, gold, silver, lead and zinc; they are classified as ferrous and non-ferrous minerals. (2) Non-metallic minerals, which do not contain metals, used in construction, ceramics, glass manufacturing, agriculture and cosmetics.

2. Differentiate between ferrous and non-ferrous minerals.

Ferrous minerals contain iron as their main component; they are

dark and magnetic, and iron is vital for industries like construction and transportation. Non-ferrous minerals do not contain iron as their main constituent; common examples are copper, aluminum, lead, zinc, tin and gold.

3. What is the difference between adit and shaft mining?

Adit mining, also known as horizontal mining or level mining, extracts minerals from underground by creating a horizontal tunnel, called an adit, into the side of a hill or mountain; adits are typically driven into the side of a slope or hill and may extend horizontally to reach the desired minerals, and the method offers cost-effectiveness and safety. Shaft mining extracts minerals from deep beneath the Earth's surface by creating vertical tunnels or shafts, drilled or excavated to provide access to underground mineral deposits; it requires specialized equipment and techniques for safety and efficiency in underground environments.

4. Write any two differences between bituminous and lignite coal.

Two differences between bituminous and lignite coal: (1) Bituminous coal is dark black in colour with high carbon content (70 percent to 86 percent), while lignite coal is brownish-black with lower carbon content (50 percent to 70 percent). (2) Bituminous coal has higher energy content, burns more efficiently producing less smoke and pollutants, and is found deeper in the Earth's crust requiring more advanced mining techniques; lignite has lower energy content, high moisture content, produces smoke and pollution, is easier to mine and is found near to the Earth.

5. What is biomass energy?

Biomass energy is derived from organic materials such as agricultural waste or wood, which are burned or converted into biofuels to produce heat or electricity. For example, agricultural waste like sugarcane biogas can be used as a fuel in biomass power plants.

Exercise: Give detailed answers of the following questions.

1. Write in detail the metallic and non-metallic minerals found in Pakistan.

Metallic minerals found in Pakistan: Copper, found in Qalat, Zhob and Chagai district of Balochistan (Saindak Copper-Gold Project) and in Dir, Chitral and Hazara in Khyber Pakhtunkhwa; used in construction, electrical wiring and electronics. Iron ore, mainly in Chiniot and Kalabagh regions of Punjab (Kalabagh is one of the largest deposits, 350 million tonnes measured reserves), also Domal Nisar (Chitral), Langrial (Abbottabad) and Chilghazi (Chaghi); used in steel production for construction and manufacturing. Chromite, primarily in Balochistan (Muslim Bagh, Chaghi, Kharan, Khuzdar) and Khyber Pakhtunkhwa (Mohmand Agency); used in stainless steel, alloys and refractory materials. Non-metallic minerals found in Pakistan: Rock salt, with extensive reserves particularly in the Khewra Salt Mine, one of the largest and oldest salt mines in the world, plus Kalabagh, Warcha and Bahadurkhail; used in the chemical industry and cooking, and known as Himalayan pink salt. Limestone, with major reserves in Daudkhel, Wah, Rohri, Hyderabad, Sibi, Dera Ghazi Khan, Kohat, Nowshera and Khuzdar; used in cement, chemical industry and construction.

Gypsum, mainly in Khewra, Dandot, Daudkhel, Qadirabad, Rohri, Kohat, Dera Ghazi Khan, Loralai and Sibi; used in construction for plastering, cement production and as a soil amendment.

2. Discuss in detail the industries related to minerals and power in Pakistan.

Mineral-related industries in Pakistan: Construction industry (limestone, granite, marble); Cement industry (limestone, clay, gypsum); Glass industry (silica sand); Steel industry (iron ore); Ceramic industry (clay, feldspar, quartz); Fertilizer industry (phosphates and sulfur); Jewelry industry (precious and semi-precious gemstones such as emeralds, rubies and sapphires). Power-related industries: power generation using fossil fuels, hydropower, wind energy and solar energy; the oil and gas industry extracts and distributes oil and natural gas; coal is used in thermal power plants; transmission and distribution deliver electricity to homes and businesses. Thermal power projects named in the unit include Port Qasim Power Plant (Karachi), Sahiwal Coal Power Plant (Sahiwal), Jamshoro Power Plant (Jamshoro), Muzaffargarh Thermal Power Plant (Muzaffargarh), Faisalabad Gas Power Plant (Faisalabad), Balloki Power Plant (Kasur), Nandipur Power Plant (Nandipur, Gujranwala), Bhikki Power Plant (Sheikhupura), Hub Power Plant (Hub, Balochistan) and Guddu Thermal Power Plant (Guddu).

3. Describe the alternate sources of energy to generate power.

Alternative sources of energy to generate power: Solar energy: solar panels or photovoltaic cells convert sunlight into electricity

without harmful emissions; the Quaid-e-Azam Solar Power Park has been installed in Bahawalpur district of Punjab. Wind energy: wind turbines harness wind to generate electricity; wind farms can produce large amounts of power, and wind power plants can be installed in Sindh Province and in Lasbela and Gwadar districts of Balochistan. Hydropower: the force of flowing or falling water drives turbines to produce electricity; Pakistan has several hydropower projects such as the Tarbela Dam and Mangla Dam. Biomass energy: derived from organic materials such as agricultural waste or wood, burned or converted into biofuels for heat or electricity, for example sugarcane biogas. These sources are renewable and sustainable, reduce reliance on fossil fuels, and offer cleaner and more environment friendly energy.

4. What are the risk factors associated with coal mining?

Risk factors associated with coal mining: cave-ins, explosions and equipment failures can result in accidents and injuries to miners; exposure to harmful substances such as coal dust, silica and toxic gases can lead to respiratory diseases and black lung disease; coal mining contributes to land degradation, deforestation, water pollution and the release of greenhouse gases, impacting the environment and contributing to climate change; underground mining can cause mine collapses, jeopardizing miners' lives and damaging infrastructure; accumulation of methane gas, a byproduct of coal, can lead to explosions if not properly controlled and monitored; miners face physical strain working in challenging conditions with heavy loads; and coal mines are susceptible to fires that release toxic fumes, posing challenges for extinguishing and

resulting in environmental and health hazards. Strict safety measures, adequate training and regular monitoring are needed to mitigate these risks.

Guidance (Teacher note, page 93): investigate how rare minerals and power resources can help boost exports, and suggest ways to better utilize resources to enhance job creation, industrial development and overall economic growth in Pakistan. This is a research task; the book fixes no single written answer for it. The book's glossary (pages 96-97) carries no exercises and is not reprinted.